Adaptive (weighted) lasso in GCLMs

0 Setting (DDK §§2–3)

Data and model. $X_1, \dots, X_n \in \mathbb{R}^p$ are i.i.d. $N(0, \Sigma^*)$ with $M^*\Sigma^* + \Sigma^* M^{*\top} + C = 0$, where $M^* \in \mathrm{Stab}_p$ and $C \in PD_p$ is known. $\hat{\Sigma} = \frac{1}{n}\sum_i X_i X_i^\top$.

Design and Gram matrix. For symmetric Σ:

$$A(\Sigma) = (\Sigma \otimes I_p) + (I_p \otimes \Sigma)K^{(p,p)}, \qquad \Gamma(\Sigma) = A(\Sigma)^\top A(\Sigma), \qquad g(\Sigma) = -A(\Sigma)^\top \mathrm{vec}(C),$$

so that

$$\tfrac{1}{2}\|M\Sigma + \Sigma M^\top + C\|_F^2 = \tfrac{1}{2}\,\mathrm{vec}(M)^\top \Gamma(\Sigma)\,\mathrm{vec}(M) - g(\Sigma)^\top \mathrm{vec}(M) + \tfrac{1}{2}\|C\|_F^2.$$

Shorthand. $\hat{\Gamma} = \Gamma(\hat{\Sigma})$, $\Gamma^* = \Gamma(\Sigma^*)$, $\hat{g}$, g^*, $\Delta_\Gamma = \hat{\Gamma} - \Gamma^*$, $\Delta_g = \hat{g} - g^*$. Indices are pairs (j,k); $m = \mathrm{vec}(M)$, $m^* = \mathrm{vec}(M^*)$.

Support. $S = \{(j,k) : M^*_{jk} \neq 0\}$, $d = |S|$; Since M^* solves the Lyapunov equation, $\Gamma^* m^* = g^*$.

Constants (DDK).

$$c_{\Gamma^*} = \|\!|(\Gamma^*_{SS})^{-1}|\!\|_\infty, \qquad c_{M^*} = \|m^*\|_\infty,$$
$$P := \Gamma^*_{S^cS}(\Gamma^*_{SS})^{-1}, \qquad \rho := \|\!|P|\!\|_\infty, \qquad \beta_{\min} := \min_{(j,k)\in S} |M^*_{jk}|$$

1 Estimator, weights, weighted irrepresentability constant

Pilot and weights

- The random pilot $\hat{M}^0$ is the lasso (diagonals penalized), at the smallest λ of its grid.
- With $\hat{\mu} := \max_{(k,l)} |\hat{M}^0_{kl}|$, the maximum over all entries including the diagonal,

$$\hat{w}_{jk} = \frac{\hat{\mu}}{|\hat{M}^0_{jk}|} \quad \text{if } \hat{M}^0_{jk} \neq 0, \qquad \hat{w}_{jk} = \infty \ \text{ (entry excluded, stays 0)} \quad \text{if } \hat{M}^0_{jk} = 0.$$

- All finite weights are ≥ 1, with equality at the largest pilot entry.

Estimator (weighted direct Lyapunov lasso, all entries penalised):

$$\begin{aligned} \hat{M} &\in \operatorname*{arg\,min}_{M\in\mathbb{R}^{p\times p}} \ \tfrac{1}{2}\|M\hat{\Sigma} + \hat{\Sigma}M^\top + C\|_F^2 + \lambda \sum_{j,k} \hat{w}_{jk}|M_{jk}| \\ &= \operatorname*{arg\,min}_{m} \ \tfrac{1}{2} m^\top \hat{\Gamma} m - \hat{g}^\top m + \lambda \|\hat{W} m\|_1, \qquad \hat{W} = \mathrm{diag}(\hat{w}). \end{aligned} \tag{1.1}$$

An infinite weight means the constraint $m_j = 0$.

Scale invariance. Replacing $(\lambda, \hat{w})$ by $(\lambda/c, c\hat{w})$ leaves the objective unchanged. The products $\lambda\hat{w}_j$ matter.

Quantities. Define

$$\underline{w} := \min_{j \in S^c} \hat{w}_j, \qquad \bar{w}_S := \max_{k \in S} \hat{w}_k < \infty, \qquad r := \frac{\bar{w}_S}{\underline{w}} = \frac{\max_{j \in S^c} |\hat{M}^0_j|}{\min_{k \in S} |\hat{M}^0_k|},$$

- $\bar{w}_S < \infty$ requires the pilot to have no zero on S, a necessary condition for recovery. In particular, all infinite weights lie in S^c.

Weighted irrepresentability constant (Zhou, van de Geer & Bühlmann 2009, Def. 8.1):

$$\rho_w := \max_{j \in S^c} \frac{1}{\hat{w}_j} \sum_{k \in S} |P_{jk}|\, \hat{w}_k \quad \Big(= \big|\!\big|\!\big| \hat{W}_{S^c}^{-1} P_{S^c \cdot} \hat{W}_{SS} \big|\!\big|\!\big|_\infty \Big). \tag{1.2}$$

A free constant. Fix any $\kappa \geq 1 + \rho$. The natural choice is $\kappa = 1 + \rho$. In DDK's setting, $\rho \leq 1 - \alpha$ and $\kappa = 2 - \alpha$ is admissible, and this is the choice that reproduces D.1.

2 W.1 (deterministic; analogue of DDK Theorem D.1)

Theorem W.1. *Let $M^* \in \mathrm{Stab}_p$ have support S, let $\hat{w}$ be weights with $\bar{w}_S < \infty$, and let $\kappa \geq 1 + \rho$. Assume:*

(W1) *Γ^*_{SS} is invertible;*

(W2) *(weighted irrepresentability) $\rho_w \leq 1 - \alpha$ for some $\alpha \in (0, 1]$;*

(W3) *$\big|\!\big|\!\big|(\Delta_\Gamma)_{\cdot S}\big|\!\big|\!\big|_\infty \leq \varepsilon_1$ and $\|\Delta_g\|_\infty \leq \varepsilon_2$, with*

$$\varepsilon_1 \leq \frac{\alpha}{3 c_{\Gamma^*}(\kappa r + \alpha)};$$

(W4) *the effective penalty off the support satisfies*

$$\lambda\, \underline{w} > \frac{3\kappa}{\alpha} \max\{c_{M^*}\varepsilon_1,\ \varepsilon_2\}.$$

Then:

(a) *The minimiser $\hat{M}$ of (1.1) is unique, its support is contained in the true support ($\hat{S} \subseteq S$), and*

$$\begin{aligned} \|\hat{M} - M^*\|_\infty \leq b_w &:= \frac{c_{\Gamma^*}}{1 - c_{\Gamma^*}\varepsilon_1}\big(c_{M^*}\varepsilon_1 + \varepsilon_2 + \lambda \bar{w}_S\big) \\ &< c_{\Gamma^*}\, \lambda\Big(\bar{w}_S + \frac{\alpha}{\kappa}\, \underline{w}\Big). \end{aligned} \tag{2.1}$$

The last bound is informative only when $\underline{w} < \infty$.

(b) *If moreover $\beta_{\min} = \min_{(j,k) \in S} |M^*_{jk}| > b_w$, then $\hat{S} = S$ and $\operatorname{sign}(\hat{M}_{jk}) = \operatorname{sign}(M^*_{jk})$ for all $(j, k) \in S$.*

Reduction to DDK Theorem D.1. Take $\hat{w} \equiv 1$ and $\kappa = 2 - \alpha$; this needs $\rho \le 1 - \alpha$. Then:

- (W2) is DDK's (D.1), $\rho \le 1 - \alpha$;
- (W3) is DDK's $\varepsilon_1 \le \alpha/(6c_{\Gamma^*})$, since $3c_{\Gamma^*}(\kappa \cdot 1 + \alpha) = 6c_{\Gamma^*}$;
- (W4) is DDK's $\lambda > \frac{3(2-\alpha)}{\alpha} \max\{c_{M^*}\varepsilon_1, \varepsilon_2\}$;
- the right-hand bound in (2.1) is $c_{\Gamma^*}\lambda(1 + \frac{\alpha}{2-\alpha}) = \frac{2c_{\Gamma^*}\lambda}{2-\alpha}$, which is DDK's bound;
- DDK's beta-min $\beta_{\min} > 2c_{\Gamma^*}\lambda/(2-\alpha)$ implies the one in (b).

3 Proof of W.1

KKT conditions Problem (1.1) is convex, since $\hat{\Gamma} \succeq 0$.

$$\begin{aligned} &\big(\hat{\Gamma}\hat{m} - \hat{g}\big)_j + \lambda\, \hat{w}_j \hat{z}_j = 0 \quad \text{(for every } j \text{ with } \hat{w}_j < \infty), \\ &\hat{z}_j = \operatorname{sign}(\hat{m}_j) \text{ if } \hat{m}_j \neq 0, \qquad \hat{z}_j \in [-1, 1] \text{ if } \hat{m}_j = 0. \end{aligned} \tag{3.1}$$

Block form. Since $\Gamma^* m^* = g^*$, we have $\hat{\Gamma}\hat{m} - \hat{g} = \Gamma^*(\hat{m} - m^*) + \Delta_\Gamma \hat{m} - \Delta_g$. With $\Delta_M := \hat{m} - m^*$, the KKT conditions (3.1) read

$$\begin{bmatrix} \Gamma^*_{SS} & \Gamma^*_{SS^c} \\ \Gamma^*_{S^cS} & \Gamma^*_{S^cS^c} \end{bmatrix} \begin{bmatrix} (\Delta_M)_S \\ (\Delta_M)_{S^c} \end{bmatrix} + \begin{bmatrix} (\Delta_\Gamma)_{SS} & (\Delta_\Gamma)_{SS^c} \\ (\Delta_\Gamma)_{S^cS} & (\Delta_\Gamma)_{S^cS^c} \end{bmatrix} \begin{bmatrix} \hat{m}_S \\ \hat{m}_{S^c} \end{bmatrix} - \begin{bmatrix} (\Delta_g)_S \\ (\Delta_g)_{S^c} \end{bmatrix} + \lambda \begin{bmatrix} (\hat{w} \circ \hat{z})_S \\ (\hat{w} \circ \hat{z})_{S^c} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

where the S^c-rows are taken only for j with $\hat{w}_j < \infty$.

▶ The PDW constructs a pair $(\tilde{m}, \tilde{z})$ satisfying (3.1) with $\tilde{m}_{S^c} = 0$.

Step 1 (restricted problem) Let

$$\tilde{m} = \operatorname*{arg\,min}_{m:\ m_{S^c}=0} \tfrac{1}{2} m^\top \hat{\Gamma} m - \hat{g}^\top m + \lambda \|\hat{W} m\|_1. \tag{3.2}$$

The restricted objective is strictly convex (as $\hat{\Gamma}$ positive semidefinite and invertible according to DDK) and $\tilde{m}$ is unique.

At $\tilde{m}$, where $\tilde{m}_{S^c} = 0$, the S-rows give

$$(\hat{\Gamma}\tilde{m} - \hat{g})_S + \lambda(\hat{w} \circ \tilde{z})_S = \hat{\Gamma}_{SS}\tilde{m}_S - \hat{g}_S + \lambda(\hat{w} \circ \tilde{z})_S = 0. \tag{3.3}$$

Error bound:

Since $m^*_{S^c} = 0$ and $\Gamma^*_{SS} m^*_S - g^*_S = 0$, we have

$$\hat{\Gamma}_{SS} m^*_S - \hat{g}_S = \underbrace{\Gamma^*_{SS} m^*_S - g^*_S}_{=0} + (\Delta_\Gamma)_{SS} m^*_S - (\Delta_g)_S = (\Delta_\Gamma)_{SS} m^*_S - (\Delta_g)_S.$$

Subtracting this from (3.3) gives

$$\hat{\Gamma}_{SS} \underbrace{(\tilde{m}_S - m^*_S)}_{=\tilde{\Delta}_S} + \lambda(\hat{w} \circ \tilde{z})_S = -((\Delta_\Gamma)_{SS} m^*_S - (\Delta_g)_S)$$

As $\hat{\Gamma}_{SS}$ invertible (shown in DDK):

$$\tilde{\Delta}_S := \tilde{m}_S - m^*_S = -(\hat{\Gamma}_{SS})^{-1} \underbrace{\Big((\Delta_\Gamma)_{SS} m^*_S - (\Delta_g)_S + \lambda(\hat{w} \circ \tilde{z})_S\Big)}_{=:v}.$$

Then: $\|\tilde{\Delta}_S\|_\infty \le |||(\hat{\Gamma}_{SS})^{-1}|||_\infty \|v\|_\infty$.

For v,

$$\begin{aligned}
\|v\|_\infty &\le \|(\Delta_\Gamma)_{SS} m^*_S\|_\infty + \|(\Delta_g)_S\|_\infty + \lambda\|(\hat{w} \circ \tilde{z})_S\|_\infty && \text{(triangle inequality)} \\
&\le |||(\Delta_\Gamma)_{\cdot S}|||_\infty \, c_{M^*} + \|\Delta_g\|_\infty + \lambda \bar{w}_S && (\|m^*_S\|_\infty \le c_{M^*},\ |\tilde{z}_k| \le 1) \\
&\le c_{M^*}\varepsilon_1 + \varepsilon_2 + \lambda \bar{w}_S && \text{(W3)}.
\end{aligned}$$

For the inverse, write $\hat{\Gamma}_{SS} = \Gamma^*_{SS} + (\Delta_\Gamma)_{SS} = \Gamma^*_{SS}(I + X)$ with $X := (\Gamma^*_{SS})^{-1}(\Delta_\Gamma)_{SS}$. Then

$$\begin{aligned}
|||X|||_\infty &\le |||(\Gamma^*_{SS})^{-1}|||_\infty \, |||(\Delta_\Gamma)_{SS}|||_\infty && \text{(submultiplicativity)} \\
&\le c_{\Gamma^*}\varepsilon_1 && \text{(W3)} \\
&\le \frac{\alpha}{3(\kappa r + \alpha)} \le \frac{1}{3} && \text{(W3)}.
\end{aligned}$$

So the Neumann series $(I + X)^{-1} = \sum_{k \ge 0} (-X)^k$ converges, $\hat{\Gamma}_{SS}$ is invertible, and

$$\begin{aligned}
|||(\hat{\Gamma}_{SS})^{-1}|||_\infty &= |||(I + X)^{-1}(\Gamma^*_{SS})^{-1}|||_\infty \\
&\le |||(I + X)^{-1}|||_\infty \, c_{\Gamma^*} && \text{(submultiplicativity)} \\
&\le \sum_{k \ge 0} |||X|||_\infty^k \, c_{\Gamma^*} = \frac{c_{\Gamma^*}}{1 - |||X|||_\infty} && \text{(Neumann series)} \\
&\le \frac{c_{\Gamma^*}}{1 - c_{\Gamma^*}\varepsilon_1} && (|||X|||_\infty \le c_{\Gamma^*}\varepsilon_1). \qquad (3.4)
\end{aligned}$$

Multiplying the two bounds gives

$$\|\tilde{\Delta}_S\|_\infty \le \frac{c_{\Gamma^*}}{1 - c_{\Gamma^*}\varepsilon_1}\big(c_{M^*}\varepsilon_1 + \varepsilon_2 + \lambda \bar{w}_S\big) =: b_w. \qquad (3.5)$$

Converting the bound into λ:

By (W4), $c_{M^*}\varepsilon_1 + \varepsilon_2 \le 2\max\{c_{M^*}\varepsilon_1, \varepsilon_2\} < \frac{2\alpha}{3\kappa}\lambda\underline{w}$. By (W3), $\frac{1}{1 - c_{\Gamma^*}\varepsilon_1} \le \frac{3(\kappa r + \alpha)}{3\kappa r + 2\alpha}$. Hence

$$\begin{aligned}
b_w &< c_{\Gamma^*} \cdot \frac{3(\kappa r + \alpha)}{3\kappa r + 2\alpha} \cdot \lambda\underline{w}\Big(r + \frac{2\alpha}{3\kappa}\Big) \\
&= c_{\Gamma^*}\lambda\underline{w}\,\frac{\kappa r + \alpha}{\kappa} = c_{\Gamma^*}\lambda\Big(\bar{w}_S + \frac{\alpha}{\kappa}\underline{w}\Big).
\end{aligned} \qquad (3.6)$$

▶ This is (D.4). With $\hat{w} \equiv 1$ and $\kappa = 2 - \alpha$ the right-hand side is DDK's $\frac{6c_{\Gamma^*}}{6-\alpha} \cdot \frac{6-\alpha}{3(2-\alpha)}\lambda = \frac{2c_{\Gamma^*}\lambda}{2-\alpha}$.

Step 2 (subgradient on S). Take $\tilde{z}_S$ from (3.3). By construction it is a subgradient of $\|\cdot\|_1$ at $\tilde{m}_S$.

Step 3 (dual vector on S^c). Evaluate the block form at $(\tilde{m}, \tilde{z})$. Since $\tilde{m}_{S^c} = m^*_{S^c} = 0$, we have $(\Delta_M)_{S^c} = 0$ and $(\Delta_M)_S = \tilde{\Delta}_S$:

$$\begin{bmatrix} \Gamma^*_{SS} & \Gamma^*_{SS^c} \\ \Gamma^*_{S^cS} & \Gamma^*_{S^cS^c} \end{bmatrix} \begin{bmatrix} \tilde{\Delta}_S \\ 0 \end{bmatrix} + \begin{bmatrix} (\Delta_\Gamma)_{SS} & (\Delta_\Gamma)_{SS^c} \\ (\Delta_\Gamma)_{S^cS} & (\Delta_\Gamma)_{S^cS^c} \end{bmatrix} \begin{bmatrix} \tilde{m}_S \\ 0 \end{bmatrix} - \begin{bmatrix} (\Delta_g)_S \\ (\Delta_g)_{S^c} \end{bmatrix} + \lambda \begin{bmatrix} (\hat{w} \circ \tilde{z})_S \\ (\hat{w} \circ \tilde{z})_{S^c} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The zeros remove the S^c-columns of Γ^* and Δ_Γ. Collect the noise in

$$u := (\Delta_\Gamma)_{\cdot S}\tilde{m}_S - \Delta_g, \qquad \text{i.e.} \quad u_S = (\Delta_\Gamma)_{SS}\tilde{m}_S - (\Delta_g)_S, \quad u_{S^c} = (\Delta_\Gamma)_{S^cS}\tilde{m}_S - (\Delta_g)_{S^c}.$$

The two block rows read

$$\begin{aligned} \Gamma^*_{SS}\tilde{\Delta}_S + u_S + \lambda(\hat{w} \circ \tilde{z})_S = 0 \qquad & (S\text{-rows}), \\ \Gamma^*_{S^cS}\tilde{\Delta}_S + u_{S^c} + \lambda(\hat{w} \circ \tilde{z})_{S^c} = 0 \qquad & (S^c\text{-rows with } \hat{w}_j < \infty). \end{aligned}$$

Solving the S-rows gives $\tilde{\Delta}_S = -(\Gamma^*_{SS})^{-1}(u_S + \lambda(\hat{w} \circ \tilde{z})_S)$.

Define $\tilde{z}_j$, for $j \in S^c$ with $\hat{w}_j < \infty$, by the S^c-rows, solved for the $\tilde{z}$-term. Then insert $\tilde{\Delta}_S$ and write $P = \Gamma^*_{S^cS}(\Gamma^*_{SS})^{-1}$:

$$\begin{aligned} \lambda(\hat{w} \circ \tilde{z})_{S^c} &= -\Gamma^*_{S^cS}\tilde{\Delta}_S - u_{S^c} && (S^c\text{-rows}) \\ &= \Gamma^*_{S^cS}(\Gamma^*_{SS})^{-1}(u_S + \lambda(\hat{w} \circ \tilde{z})_S) - u_{S^c} && (\text{insert } \tilde{\Delta}_S) \\ &= P(u_S + \lambda(\hat{w} \circ \tilde{z})_S) - u_{S^c} && (\text{definition of } P) \\ &= Pu_S - u_{S^c} + \lambda P(\hat{w} \circ \tilde{z})_S && (\text{expand}), \end{aligned}$$

and dividing row j by $\lambda\hat{w}_j$,

$$\tilde{z}_j = \underbrace{\frac{1}{\lambda\hat{w}_j}\big[Pu_S - u_{S^c}\big]_j}_{\text{noise}} + \underbrace{\frac{1}{\hat{w}_j}\big[P(\hat{w} \circ \tilde{z})_S\big]_j}_{\text{deterministic}}. \tag{3.7}$$

▶ This is (D.5) with $\operatorname{sign}(\operatorname{vec}(\tilde{M})_S)$ replaced by $(\hat{w} \circ \tilde{z})_S$ and row j divided by $\hat{w}_j$.

Deterministic term. Since $|\tilde{z}_k| \le 1$,

$$\Big|\frac{1}{\hat{w}_j}\big[P(\hat{w} \circ \tilde{z})_S\big]_j\Big| \le \frac{1}{\hat{w}_j}\sum_{k \in S}|P_{jk}|\hat{w}_k \le \rho_w \le 1 - \alpha \qquad \text{by (W2)}. \tag{3.8}$$

▶ DDK bound this term by $\rho \le 1 - \alpha$. This is the only place where the weighted condition enters.

Noise term.

$$\begin{aligned} |\text{noise}_j| &\le \frac{1}{\lambda\hat{w}_j}(\rho\|u_S\|_\infty + \|u_{S^c}\|_\infty) && (\|\!|P|\!\|_\infty = \rho) \\ &\le \frac{\kappa}{\lambda\underline{w}}\|u\|_\infty && (1 + \rho \le \kappa,\ \hat{w}_j \ge \underline{w}) \\ &\le \frac{\kappa}{\lambda\underline{w}}(\varepsilon_1\|\tilde{m}_S\|_\infty + \varepsilon_2) && \text{Def. of } u, \text{ (W3)} \\ &\le \underbrace{\frac{\kappa\, c_{M^*}\varepsilon_1}{\lambda\underline{w}}}_{G_1} + \underbrace{\frac{\kappa\,\varepsilon_1\|\tilde{\Delta}_S\|_\infty}{\lambda\underline{w}}}_{G_2} + \underbrace{\frac{\kappa\,\varepsilon_2}{\lambda\underline{w}}}_{G_3} && (\|\tilde{m}_S\|_\infty \le c_{M^*} + \|\tilde{\Delta}_S\|_\infty). \end{aligned} \tag{3.9}$$

▶ These are DDK's G_1, G_2, G_3 (pp. 532–533), with $(2-\alpha)/\lambda$ replaced by $\kappa/(\lambda\underline{w})$. Each is below $\alpha/3$:

- $G_1, G_3 < \alpha/3$ by (W4);
- by (3.5)–(3.6), $G_2 < \frac{\kappa\varepsilon_1}{\lambda\underline{w}}\, c_{\Gamma^*}\lambda\underline{w}\,\frac{\kappa r+\alpha}{\kappa} = c_{\Gamma^*}\varepsilon_1(\kappa r+\alpha) \le \alpha/3$ by (W3).

With (3.8),

$$\max_{j\in S^c,\ \hat{w}_j<\infty} |\tilde{z}_j| < (1-\alpha) + \tfrac{\alpha}{3} + \tfrac{\alpha}{3} + \tfrac{\alpha}{3} = 1. \tag{3.10}$$

If $\hat{w}_j = \infty$ for every $j \in S^c$ (the pilot is zero on all of S^c), Step 3 is empty: every S^c-coordinate is fixed at 0 by the constraint, so there is no $\tilde{z}_j$ to bound and $\hat{m}_{S^c} = 0$ holds automatically.

Step 4 (the witness is the unique minimiser; $\hat{S} \subseteq S$). The pair $(\tilde{m}, \tilde{z})$ satisfies (3.1):

- the S-rows by (3.3) and the rows $j \in S^c$ with $\hat{w}_j < \infty$ by construction;
- $\tilde{m}_j = 0$ wherever $\hat{w}_j = \infty$, since all infinite weights lie in S^c;
- $\tilde{z}$ is a valid subgradient, since $|\tilde{z}_j| < 1$ at the zero coordinates $j \in S^c$ with $\hat{w}_j < \infty$ by (3.10), and $\tilde{z}_S$ comes from Step 2.

So $\tilde{m}$ is a minimiser.

Every minimiser has support in S (Wainwright 2009, Lemma 1, cited by DDK on p. 532). Strict dual feasibility implies hat $\hat{S} \subseteq S$.

Uniqueness. Every minimiser lives on $\{m_{S^c} = 0\}$, where the objective is strictly convex (Step 1). Part (a) follows from (3.5)–(3.6). $\square_{(a)}$

Part (b). For $(j,k) \in S$, (a) gives $|\hat{M}_{jk} - M^*_{jk}| \le b_w < \beta_{\min} \le |M^*_{jk}|$. Hence $\hat{M}_{jk} \neq 0$ with the sign of M^*_{jk}, so $S \subseteq \hat{S}$, and with (a), $\hat{S} = S$.

4 Theorem W.2 (probabilistic; the analogue of DDK Theorem 2)

DDK's tools, used unchanged.

- Lemma E.6: $|||\Delta_\Sigma|||_2 < \min\{\varepsilon_1/(\sqrt{d}(4+8c_{\Sigma^*})),\ \varepsilon_2/(2c_C)\}$ implies $|||(\Delta_\Gamma)_{\cdot S}|||_\infty < \varepsilon_1$ and $\|\Delta_g\|_\infty < \varepsilon_2$, with $c_{\Sigma^*} = |||\Sigma^*|||_2$ and $c_C = \|\operatorname{vec}(C)\|_2$.
- Corollary E.5 (from Wainwright 2019, Thm 6.5) bounds $|||\Delta_\Sigma|||_2$.
- Neither involves the penalty. As in DDK,

$$\tilde{c} = \max\Big\{\frac{4\max\{1,c^2_{\Sigma^*}\}(4+8c_{\Sigma^*})^2}{c_3},\ 16c_1^2c^2_{\Sigma^*}(4+8c_{\Sigma^*})^2,\ \frac{16\max\{1,c^2_{\Sigma^*}\}c_C^2}{c_3},\ 64c_1^2c^2_{\Sigma^*}c_C^2\Big\},$$

  with the universal constants c_1, c_2, c_3 of Theorem E.4.

Theorem W.2. *In the setting of DDK Theorem 2, let $\kappa \ge 1+\rho$, $\alpha \in (0,1]$ and $\bar{r} \ge 0$. Let $\mathcal{E}_w$ be an event on which the weights satisfy $\bar{w}_S < \infty$, $\rho_w \le 1-\alpha$ and $r \le \bar{r}$. Put*

$$\varepsilon := \sqrt{\tau_1\tilde{c}\,d\,p/n}, \qquad c_w := \frac{3c_{\Gamma^*}(\kappa\bar{r}+\alpha)}{\alpha} \qquad \Big(\textit{DDK: } c_* = \frac{6c_{\Gamma^*}}{\alpha}\Big).$$

Suppose $n > \tau_1\tilde{c}\,d\,p\max\{c_w^2, 1/4\}$ for some $\tau_1 > 1$, and that on $\mathcal{E}_w$

$$\lambda\,\underline{w} > \frac{3\kappa}{\alpha}\max\{c_{M^*}, 1\}\,\varepsilon.$$

For the implemented weights ($\hat{w} \geq 1$) it suffices that $\lambda > \frac{3\kappa}{\alpha}\max\{c_{M^*}, 1\}\varepsilon$; this is DDK's condition with $2-\alpha$ replaced by κ. Then, with probability at least $1 - c_2\exp(-\tau_1 p) - P(\mathcal{E}_w^c)$:

(a) *the minimiser of (1.1) is unique, $\hat{S} \subseteq S$, and*

$$\|\hat{M} - M^*\|_\infty \leq \frac{c_{\Gamma^*}}{1 - c_{\Gamma^*}\varepsilon}\big((c_{M^*}+1)\varepsilon + \lambda\bar{w}_S\big) < c_{\Gamma^*}\lambda\Big(\bar{w}_S + \frac{\alpha}{\kappa}\underline{w}\Big);$$

(b) *if $\beta_{\min}$ exceeds the first bound in (a), then $\hat{S} = S$ with the signs of M^*.*

Proof. DDK Appendix F, steps 1–3, with $\varepsilon_1 = \varepsilon_2 = \varepsilon$.

- **Steps 1–2 are unchanged.** $\varepsilon/(\sqrt{d}(4 + 8c_{\Sigma^*}))$ and $\varepsilon/(2c_C)$ lie in the admissible range of Corollary E.5. The upper end uses only $\max\{c_w^2, 1/4\} \geq 1/4$, the lower end only $\tilde{c}$. So with probability at least $1 - c_2\exp(-\tau_1 p)$, $\|\!|(\Delta_\Gamma)_{\cdot S}|\!\|_\infty < \varepsilon$ and $\|\Delta_g\|_\infty < \varepsilon$. Both are bounds on the same random variable $\|\!|\Delta_\Sigma|\!\|_2$, so no union bound is needed.
- **Step 3 replaces c_* by c_w.** $\varepsilon < 1/\max\{c_w, 1/2\} \leq 1/c_w = \alpha/(3c_{\Gamma^*}(\kappa\bar{r} + \alpha)) \leq \alpha/(3c_{\Gamma^*}(\kappa r + \alpha))$ on $\mathcal{E}_w$, which is (W3).
- **Conclude.** The λ-condition is (W4), since $\max\{c_{M^*}\varepsilon, \varepsilon\} = \max\{c_{M^*}, 1\}\varepsilon$. Apply Theorem W.1 on $\mathcal{E}_w \cap \{\text{concentration}\}$. □

Reading. With $\hat{w} \equiv 1$, $\kappa = 2 - \alpha$ and $\bar{r} = 1$, Theorem W.2 is DDK's Theorem 2. With a good pilot, $\bar{r} \to 0$ and one may take $\alpha \to 1$, so $c_w \to 3c_{\Gamma^*}$. The sample size then no longer depends on an irrepresentability gap: it becomes $n \gtrsim \tilde{c}\,dp \cdot 9c_{\Gamma^*}^2$, against DDK's $\tilde{c}\,dp \cdot 36c_{\Gamma^*}^2/\alpha^2$ with α the (often non-existent) lasso gap. The order $\Omega(dp)$ in (d, p) is unchanged.

5 Corollaries, necessity, and the dense-end pilot

Corollary W.3 (A pilot close to the truth)**.** *Let $\delta < \beta_{\min}/(1+\rho)$ and put*

$$\bar{r} := \frac{\delta}{\beta_{\min} - \delta}, \qquad \alpha := 1 - \rho\,\bar{r}.$$

Then $\alpha \in (0, 1]$, and on the event $\mathcal{E}_\delta := \{\|\hat{M}^0 - M^*\|_\infty \leq \delta\}$:

(i) $\hat{M}^0_k \neq 0$ for every $k \in S$; hence $\bar{w}_S < \infty$, and every infinite weight lies in S^c;

(ii) $r \leq \bar{r}$;

(iii) $\rho_w \leq \rho\,r \leq 1 - \alpha$, i.e. (W2) holds with this α.

*Consequently, **whatever the size of DDK's constant** ρ:*

(a) on $\mathcal{E}_\delta$, the conclusions of Theorem W.1 hold with this α whenever (W1), (W3) and (W4) hold; since $r \leq \bar{r}$, (W3) is implied by $\varepsilon_1 \leq \alpha/(3c_{\Gamma^*}(\kappa\bar{r} + \alpha))$;

(b) Theorem W.2 applies with $\mathcal{E}_w := \mathcal{E}_\delta$ and with this α and $\bar{r}$: under its sample-size and λ-conditions, its conclusions hold with probability at least $1 - c_2\exp(-\tau_1 p) - P(\|\hat{M}^0 - M^*\|_\infty > \delta)$.

Proof. Throughout, work on $\mathcal{E}_\delta$, so that $|\hat{M}^0_j - M^*_j| \leq \|\hat{M}^0 - M^*\|_\infty \leq \delta$ for every entry j.

Step 0: $\alpha \in (0, 1]$. Since $\rho \geq 0$, we have $\delta < \beta_{\min}/(1+\rho) \leq \beta_{\min}$, so $\beta_{\min} - \delta > 0$, $\bar{r} \geq 0$ is well defined, and $\alpha \leq 1$. Moreover, multiplying by $\beta_{\min} - \delta > 0$,

$$\alpha > 0 \iff \rho\,\frac{\delta}{\beta_{\min} - \delta} < 1 \iff \rho\,\delta < \beta_{\min} - \delta \iff (1+\rho)\,\delta < \beta_{\min},$$

and the last inequality is the assumption.

Step 1: location of the pilot entries; proof of (i). Off the support: for $j \in S^c$ we have $M^*_j = 0$, hence

$$|\hat{M}^0_j| \;=\; |\hat{M}^0_j - M^*_j| \;\le\; \delta. \tag{5.1}$$

On the support: for $k \in S$, the triangle inequality $|M^*_k| \le |M^*_k - \hat{M}^0_k| + |\hat{M}^0_k|$ gives the reverse triangle inequality

$$|\hat{M}^0_k| \;\ge\; |M^*_k| - |\hat{M}^0_k - M^*_k| \;\ge\; \beta_{\min} - \delta \;>\; 0, \tag{5.2}$$

where we used $|M^*_k| \ge \beta_{\min}$ (definition of $\beta_{\min}$ as the minimum over S), $|\hat{M}^0_k - M^*_k| \le \delta$, and $\delta < \beta_{\min}$. Hence $\hat{M}^0_k \neq 0$ for every $k \in S$, so every weight on S falls into the first case of the weight definition, $\hat{w}_k = \hat{\mu}/|\hat{M}^0_k|$, with $\hat{\mu} < \infty$ and a strictly positive denominator. Thus each $\hat{w}_k$, $k \in S$, is finite, and $\bar{w}_S = \max_{k\in S} \hat{w}_k$, a maximum of finitely many finite numbers, is finite. An infinite weight occurs only at a pilot zero, which by (5.2) is impossible on S. Hence the constraint $m_j = 0$ imposed by $\hat{w}_j = \infty$ only fixes entries with $M^*_j = 0$; no entry of the true support is excluded from the estimator.

Step 2: proof of (ii). By (5.2), $\hat{\mu} \ge |\hat{M}^0_k| \ge \beta_{\min} - \delta > 0$ for any $k \in S$. With the convention $\hat{\mu}/0 := \infty$, which matches the definition of the excluded weights, $\hat{w}_j = \hat{\mu}/|\hat{M}^0_j|$ for every entry, and $t \mapsto \hat{\mu}/t$ is decreasing on $[0, \infty)$. Hence the largest weight on S belongs to the smallest pilot entry on S, and the smallest weight on S^c belongs to the largest pilot entry on S^c:

$$\bar{w}_S = \frac{\hat{\mu}}{\min_{k\in S} |\hat{M}^0_k|}, \qquad \underline{w} = \frac{\hat{\mu}}{\max_{j\in S^c} |\hat{M}^0_j|}.$$

(If $\hat{M}^0_{S^c} = 0$, then $\underline{w} = \infty$ and $r = 0$, and (ii) is trivial.) Therefore

$$r = \frac{\bar{w}_S}{\underline{w}} = \frac{\max_{j\in S^c} |\hat{M}^0_j|}{\min_{k\in S} |\hat{M}^0_k|} \le \frac{\delta}{\beta_{\min} - \delta} = \bar{r},$$

because by (5.1) the numerator is at most δ, by (5.2) the denominator is at least $\beta_{\min} - \delta > 0$, and such a ratio increases when its numerator increases or its denominator decreases.

Step 3: proof of (iii). Let $j \in S^c$. If $\hat{w}_j = \infty$, the j-th term in the definition (1.2) of ρ_w is 0. Otherwise, using $\hat{w}_j \ge \underline{w}$, $\hat{w}_k \le \bar{w}_S$ for $k \in S$, and $\sum_{k\in S} |P_{jk}| \le |||P|||_\infty = \rho$ (maximum absolute row sum),

$$\frac{1}{\hat{w}_j} \sum_{k\in S} |P_{jk}|\, \hat{w}_k \;\le\; \frac{\bar{w}_S}{\underline{w}} \sum_{k\in S} |P_{jk}| \;\le\; r\,\rho.$$

Taking the maximum over $j \in S^c$ gives $\rho_w \le \rho\, r$. By (ii) and the definition of α, $\rho_w \le \rho\, r \le \rho\, \bar{r} = 1 - \alpha$, and $\alpha \in (0, 1]$ by Step 0. This is (W2).

Step 4: consequences. (a) By (i), the standing assumption $\bar{w}_S < \infty$ of Theorem W.1 holds, and by (iii) so does (W2). The remaining hypotheses (W1), (W3), (W4) are assumed. The right-hand side of (W3) is decreasing in r and $r \le \bar{r}$ by (ii), so $\varepsilon_1 \le \alpha/\big(3c_{\Gamma^*}(\kappa\bar{r} + \alpha)\big)$ implies (W3).

(b) Theorem W.2 requires an event on which $\bar{w}_S < \infty$, $\rho_w \le 1 - \alpha$ and $r \le \bar{r}$, for deterministic α and $\bar{r}$. By (i)–(iii), $\mathcal{E}_\delta$ is such an event, and α, $\bar{r}$ depend only on δ, $\beta_{\min}$ and ρ. Theorem W.2 therefore gives its conclusions with probability at least $1 - c_2 \exp(-\tau_1 p) - P(\mathcal{E}^c_\delta)$. □

Remark W.4.

- Since the diagonal is penalised, $\beta_{\min} = \min_{(j,k)\in S} |M^*_{jk}|$ runs over all of S, diagonal included. The diagonal entries are typically the largest in magnitude, so in practice $\beta_{\min}$ is the weakest true edge, and the tolerance $\beta_{\min}/(1 + \rho)$ is set by that edge.

- DDK need $\rho \le 1-\alpha < 1$, a property of (Σ^*, S) that no amount of data changes. Here ρ only determines how accurate the pilot must be: for every ρ the tolerance $\beta_{\min}/(1+\rho)$ is positive. The constant ρ does not disappear, however. It enters $\kappa \ge 1+\rho$ in (W3)–(W4) and $\alpha = 1-\rho\bar{r}$, and as $\delta \uparrow \beta_{\min}/(1+\rho)$ we have $\alpha \downarrow 0$, so (W3), (W4) and the sample size in Theorem W.2 become arbitrarily demanding.
- For a prescribed $\alpha \in (0,1]$, the condition $\rho\bar{r} \le 1-\alpha$ is equivalent to $\delta \le (1-\alpha)\beta_{\min}/(\rho+1-\alpha)$. The α of Corollary W.3 is the largest admissible one, and also the best choice, since (W3) and (W4) become weaker requirements as α grows.

Corollary W.5 (Sign-weighted version)**.** *Assume (W1), (W3), (W4) and $\beta_{\min} > b_w$. Replace (W2) by the weaker condition*

$$\rho_w^{\mathrm{sgn}} := \max_{j \in S^c} \frac{1}{\hat{w}_j} \Big| \sum_{k \in S} P_{jk}\, \hat{w}_k \operatorname{sign}(M^*_k) \Big| \le 1-\alpha,$$

where the sum includes the diagonal, with $\operatorname{sign}(M^*_{ii}) = -1$ in the thesis DGP. Then the conclusions of Theorem W.1 hold.

Proof. (3.5) does not use (W2). With $\beta_{\min} > b_w$ it gives $\operatorname{sign}(\tilde{m}_S) = \operatorname{sign}(m^*_S)$, hence $\tilde{z}_S = \operatorname{sign}(m^*_S)$. The deterministic term of (3.7) then equals the expression inside ρ_w^{sgn}; the rest of the proof is unchanged. □

This is the weighted form of DDK's weak condition (G.1). It is the quantity to compute per dataset; `gclmth.adaptive_conditions` returns it for an unpenalised diagonal and needs the diagonal signs added for this version.

Proposition W.6 (Necessity; the analogue of DDK Prop. G.2)**.** *Let $\lambda > 0$ be fixed and let the weights converge in probability, $\hat{w} \to w$, with $0 < w_j < \infty$ on S^c and $w < \infty$ on S. Assume $\beta_{\min} > c_{\Gamma^*} \lambda \max_{k \in S} w_k$ and*

$$\max_{j \in S^c} \frac{1}{w_j} \Big| \sum_{k \in S} P_{jk}\, w_k \operatorname{sign}(M^*_k) \Big| > 1.$$

Then $P(\hat{S} = S) \to 0$.

Proof. As in G.2. On $\{\hat{S} = S\}$, $\hat{M}$ is the restricted solution (3.2), and by (3.5) and $\Delta_\Gamma, \Delta_g \to 0$ in probability (Lemma E.6, Corollary E.5) it has the signs of M^* with probability tending to one. Its dual vector is (3.7). The noise part tends to 0, and the deterministic part tends to the displayed quantity, which exceeds 1 in some row. That violates (3.1), so $P(\hat{S} = S) \to 0$. □

Remark W.7 (The dense-end pilot when the diagonal is penalised)*.*

- As $\lambda \to 0$, and in the population, the pilot tends to the minimum-$\|M\|_1$ exact solution (all entries) of $M\Sigma^* + \Sigma^* M^\top + C = 0$ (Tibshirani 2013). Penalising the diagonal changes this basis-pursuit problem, but only in one direction.
- *Trace identity.* Every exact solution has $\operatorname{tr}(M) = -\frac{1}{2}\operatorname{tr}(C\Sigma^{*-1})$. To see this, multiply the equation by Σ^{*-1} from the right and take traces, using $\operatorname{tr}(\Sigma^* M^\top \Sigma^{*-1}) = \operatorname{tr}(M)$. Hence $\sum_i |M_{ii}| \ge \frac{1}{2}\operatorname{tr}(C\Sigma^{*-1})$, with equality if and only if all $M_{ii} \le 0$.
- *Consequence.* Suppose M^* has a non-positive diagonal (every DGP of the thesis does) and is the unique minimiser of $\|M_{\mathrm{off}}\|_1$ over exact solutions; this is the BP condition checked so far. Then for every exact $M \ne M^*$,

$$\|M\|_1 \ge \|M_{\mathrm{off}}\|_1 + \tfrac{1}{2}\operatorname{tr}(C\Sigma^{*-1}) > \|M^*_{\mathrm{off}}\|_1 + \tfrac{1}{2}\operatorname{tr}(C\Sigma^{*-1}) = \|M^*\|_1.$$

So the off-diagonal BP condition implies the penalised-diagonal one. In the population, penalising the diagonal can only help the dense-end pilot, and the BP counts computed so far are lower bounds.

- For an exact check of the penalised version, the dual certificate requires $A_S^\top h = \operatorname{sign}(m_S^*)$ on all of S, diagonal included, instead of $A_D^\top h = 0$ on the diagonal. This is a one-line change in `bp_certificate`.
- When the BP condition holds, $\hat{M}^0 \to M^*$, so $E \supseteq S^c$ in the limit, $r \to 0$, $\rho_w \to 0$, and Corollary W.3 applies.

6 What changed relative to DDK, line by line

DDK (lasso, $\hat{w} \equiv 1$)	here (weights $\hat{w}_j = \hat{\mu}/\lvert\hat{M}_j^0\rvert \geq 1$)	why
(D.1) $\rho < 1 - \alpha$	(W2) $\rho_w \leq 1 - \alpha$, with $\rho_w \leq \rho r$	the deterministic part of the dual vector (3.8) is $\hat{w}_j^{-1}[P(\hat{w} \circ \tilde{z})_S]_j$
$\varepsilon_1 \leq \alpha/(6c_{\Gamma^*})$	(W3) $\varepsilon_1 \leq \alpha/(3c_{\Gamma^*}(\kappa r + \alpha))$	G_2 carries κr: bias $\lambda \bar{w}_S$ on S against kink $\lambda \underline{w}$ on S^c
$\lambda > \frac{3(2-\alpha)}{\alpha} \max\{c_{M^*}\varepsilon_1, \varepsilon_2\}$	(W4) $\lambda \underline{w} > \frac{3\kappa}{\alpha} \max\{c_{M^*}\varepsilon_1, \varepsilon_2\}$; implied by $\lambda > \frac{3\kappa}{\alpha} \max\{\cdot\}$ since $\hat{w} \geq 1$	noise is compared with the smallest effective kink; $2 - \alpha \to \kappa \geq 1 + \rho$ because ρ may exceed 1
(D.4) $\lVert\tilde{\Delta}_M\rVert_\infty < \frac{2c_{\Gamma^*}\lambda}{2-\alpha}$	(3.5)–(3.6) $\leq b_w < c_{\Gamma^*}\lambda(\bar{w}_S + \frac{\alpha}{\kappa}\underline{w})$	bias $\lambda \to \lambda \bar{w}_S$
(D.5)	(3.7)	$\operatorname{sign}(\cdot) \to \hat{w} \circ \tilde{z}$, row j divided by $\hat{w}_j$
beta-min $> \frac{2c_{\Gamma^*}\lambda}{2-\alpha}$	$\beta_{\min} > b_w$ (all of S, diagonal included)	same role
Thm 2: $c_* = 6c_{\Gamma^*}/\alpha$	$c_w = 3c_{\Gamma^*}(\kappa\bar{r} + \alpha)/\alpha$	from (W3)
Prop. G.2	Prop. W.6	dual vector with limiting weights

Everything specific to GCLMs enters exactly where it enters in DDK: through the invertibility of Γ^*_{SS}, the constants $c_{\Gamma^*}, \rho, c_{M^*}, c_{\Sigma^*}, c_C$, and Appendix E. Invertibility of Γ^*_{SS} holds for every simple graph (Dettling et al. 2023).

7 Remarks

Numerical check (`check_W1.py`, `gclmth`; weighted lasso with the diagonal penalised and the implemented weights).

- *In-star, $p = 6$, $|w| = 0.7$, correlation scale.* DDK's constant is $\rho = 1.78$, so the lasso's condition fails. The pilot is truth $+U(-0.05, 0.05)$, giving $\rho_w = 0.066$ and $r = 0.052$. At $n = 10^6$ and 10^7, (W3) holds. With λ at (W4) plus 1 %, the weighted lasso returned exactly S, with $\lVert\hat{M} - M^*\rVert_\infty = 0.13$ and 0.057, below $b_w = 0.79$ and 0.31, which are below the right-hand side of (2.1).
- *Unweighted 3-path with ordered diagonal, raw scale ($\rho = 0.81$, DDK's case), $\kappa = 2 - \alpha$.* The right-hand side of (2.1) equals DDK's $2c_{\Gamma^*}\lambda/(2 - \alpha)$ to machine precision. $\hat{S} \subseteq S$ held, and part (b) was not reached because λ is large relative to $\beta_{\min} = 0.2$.

The required n is large because DDK's ε_1 (row sums of Δ_Γ over the support columns) is a conservative quantity. The theorems are sufficient conditions, as in DDK.

Unpenalised diagonal (the default runs, waves 1–9). Set $\hat{w} = 0$ on the diagonal D, take the maximum defining $\hat{\mu}$ over off-diagonal entries, replace S by $S \setminus D$ in $\bar{w}_S$, (1.2), Corollary W.5 and the beta-min, and read the diagonal rows of (3.1) as $(\hat{\Gamma}\hat{m} - \hat{g})_D = 0$. Nothing else changes; the proof is the one above.

Points in DDK to be aware of when citing.

1. g should carry the transpose (§0).
2. D.1's λ-condition is printed as $\max\{c_{M^*}, \varepsilon_1, \varepsilon_2\}$; the proof uses $\max\{c_{M^*}\varepsilon_1, \varepsilon_2\}$ (G_1, G_3 on pp. 532–533), and Theorem 2 uses $c_{M^*}\varepsilon$ (implicitly $c_{M^*} \geq 1$).
3. The beta-min displays index $(j, k) \in S$ with $1 \leq j < k$, inherited from the symmetric setting of Lin et al.; for a drift matrix it must run over all of S.
4. The block form on p. 531 writes $+\Delta_g$; with $\Delta_g = \hat{g} - g^*$ the sign is $-\Delta_g$. Only norms enter, so nothing depends on it.